Student 14 **Jeyakannan L** 192525376RD 5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 3.33 **TC: 2/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);

    }


    public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    Student s = new Student();
    s.rollNo=sc.nextInt();
    sc.nextLine();
    s.name=sc.nextLine();
    s.marks=sc.nextDouble();
    s.displayDetails();
    sc.close();

}
}
```

INPUT

```
101
jk
75.93
```

OUTPUT

```
Roll No: 101
Name: jk
Marks: 75.93
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    public void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    public void setName(String name){
        this.name=name;

    }
    public void setMarks(double marks){
        this.marks=marks;

    }
    public void displayDetails(){
        System.out.println("Roll No: "+ rollNo);
        System.out.println("Name: "+ name);
        System.out.println("Marks: "+ marks);
    }



    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s = new Student();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
101
rahul
89.5
```

OUTPUT

```
Roll No: 101
Name: rahul
Marks: 89.5
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo = 101;
        name = "Rahul";
        marks=85.5;
    }
    Student(int rollNo, String name, double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;

    }
    void displayDetails(){
        System.out.println("Roll No: "+ rollNo);
        System.out.println("Name: "+ name);
        System.out.println("Marks: "+marks);

    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s1=new Student();
        s1.displayDetails();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student s2=new Student(rollNo,name,marks);
        s2.displayDetails();
        sc.close();



    }
}
```

INPUT

```
102
Arun
92.5
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 102
Name: Arun
Marks: 92.5
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 0.00 **TC: 0/3 passed**

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
     String empName;
    double basicSalary;

    void getEmployeeDetails(int empId,String empName,double basicSalary){
        this.empId=empId;
        this.empName=empName;
        this.basicSalary=basicSalary;
   }
    public static void main(String[] args){
           Scanner sc=new Scanner(System.in);
           int empId=sc.nextInt();
           sc.nextLine();
           String empName=sc.nextLine();
           double basicSalary=sc.nextDouble();
           Salary s1=new Salary();
           s1.getEmployeeDetails(empId,empName,basicSalary);
           s1.calculateSalary();
           s1.displayDetails();
           sc.close();
       }

    }




class Salary extends Employee{
    double hra, da , netSalary;
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        netSalary=hra+da;


    }


    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Net Salary: "+netSalary);


    }
}
```

INPUT

```
101
ram
50000
```

OUTPUT

```
Employee ID: 101
Employee Name: ram
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Net Salary: 15000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 0.00 **TC: 0/3 passed**

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    void getEmployeeDetails(int empId,String empName){
        this.empId=empId;
        this.empName = empName;


    }
public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int empId=sc.nextInt();
    sc.nextLine();
    String empName=sc.nextLine();
    double basicSalary=sc.nextDouble();
    GrossSalary g=new GrossSalary();
    g.getEmployeeDetails(empId,empName);
    g.getBasicSalary(basicSalary);
    g.calculateSalary();
    g.displayDetails();
    sc.close();


}
}
class Salary extends Employee {
    double basicSalary;
    void getBasicSalary(double basicSalary){
        this.basicSalary=basicSalary;
    }
}
class GrossSalary extends Salary{
    Double hra,da,grossSalary;
    void calculateSalary(){
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        grossSalary=basicSalary+hra+da;

    }
    void displayDetails(){
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic salary: "+basicSalary);
        System.out.println("HRA : "+hra);
        System.out.println("DA: "+da);
        System.out.println("Gross Salary: "+grossSalary);

    }
}
```

INPUT

```
101
ram
50000
```

OUTPUT

```
Employee ID: 101
Employee Name: ram
Basic salary: 50000.0
HRA : 10000.0
DA: 5000.0
Gross Salary: 65000.0
```